

## Food Truck · answer reference

Classroom constants: batch 20; ingredient line costs 10, 18, 40, 10, 12 → batch \$90 → variable \$4.50. Fixed \$450. Service  $32 \times 3 = 96$ .

At  $q=100$ : quantities 10 kg, 15 kg, 10 kg, 2.5 L, 100 packages; prepared lines 50, 90, 200, 50, 60; variable total \$450.

| p  | Contribution | Markup | Crossing        | First non-loss / profit |
|----|--------------|--------|-----------------|-------------------------|
| 9  | 4.50         | 100.0% | 100             | 100 / 101               |
| 12 | 7.50         | 166.7% | 60              | 60 / 61                 |
| 15 | 10.50        | 233.3% | $\frac{300}{7}$ | 43 / 43                 |

$R=px$ ;  $C=450+4.5x$ ;  $P=(p-4.5)x-450$ . Accept equivalent forms, e.g.  $12x-(450+4.5x)$ . A right value at one  $x$  is insufficient. Markup divides by cost, not selling price.

| p  | q   | s  | Revenue | Expense | Profit  | Waste      | Business outcome    |
|----|-----|----|---------|---------|---------|------------|---------------------|
| 9  | 60  | 60 | 540.00  | 720.00  | -180.00 | 0 / 0.0%   | Business rethink    |
| 9  | 80  | 80 | 720.00  | 810.00  | -90.00  | 0 / 0.0%   | Business rethink    |
| 9  | 100 | 96 | 864.00  | 900.00  | -36.00  | 4 / 4.0%   | Business rethink    |
| 9  | 120 | 96 | 864.00  | 990.00  | -126.00 | 24 / 20.0% | Business rethink    |
| 12 | 60  | 60 | 720.00  | 720.00  | 0.00    | 0 / 0.0%   | Viable first launch |
| 12 | 80  | 80 | 960.00  | 810.00  | 150.00  | 0 / 0.0%   | Festival success    |
| 12 | 100 | 90 | 1080.00 | 900.00  | 180.00  | 10 / 10.0% | Festival success    |
| 12 | 120 | 90 | 1080.00 | 990.00  | 90.00   | 30 / 25.0% | Viable first launch |
| 15 | 60  | 60 | 900.00  | 720.00  | 180.00  | 0 / 0.0%   | Festival success    |
| 15 | 80  | 60 | 900.00  | 810.00  | 90.00   | 20 / 25.0% | Viable first launch |
| 15 | 100 | 60 | 900.00  | 900.00  | 0.00    | 40 / 40.0% | Viable first launch |
| 15 | 120 | 60 | 900.00  | 990.00  | -90.00  | 60 / 50.0% | Business rethink    |

Outcomes assume complete current evidence with no bypass. \$9 break-even 100 cannot be reached with service capacity 96; correct losses must not be rejected.

## Representations, revision and review

At  $p=12$ ,  $R(0,60,100)=0,720,1200$ ;  $C=450,720,900$ ; crossing  $(60,720)$ . At  $p=15$  crossing  $(\frac{300}{7}, \frac{4500}{7})$ . Exact fraction required.

Launch  $s=\min(q,D,96)$ ; expense  $450+4.5q$ . At  $p=12, q=100:90$  sold, revenue 1080, expense 900, profit 180, waste 10, sell-through 90%.  $P(90)=225$  overstates by 45.

$A=(12,100), B=(12,80)$ : profit change  $-30$ , waste change  $-10$ .  $B=(15,60)$ : profit change 0, waste change  $-10$ . Either recommendation may be defensible.

## Individual transfer

A:  $P(x)=9x-360$ ; equality 40; first positive 41. B: revenue 560, expense 480, profit 80, waste 14, sell-through 80%, prepared stock basis.

## Accepted forms

$4.5=4.50=9/2$ . Linear expanded/factored equivalents pass. \$150 or 150 USD in a total-money field; USD/portion is not a total. Counts require exact integer values; percentages rounded half-up to the stated 0.1%.

## Misconceptions to inspect

Revenue is not profit. Fixed fees are paid once. Unsold prepared food still costs money. Break-even is not always the first profitable sale. Demand is not capacity. Check all tied limits.

## Teacher review

Record reasoning or oral/paper transfer with an alias and observation. Structural submission does not grade prose. Teacher bypass remains provisional. First attempts, assistance, carried-forward and imported evidence do not establish independent mastery.

## Classroom operation

Pairs swap roles after lessons 1/2 and export progress. Test the actual iPad, school network and printer. Print A4/Letter portrait 100%, browser headers/footers off. Keep student reports, answer reference and transfer slips separate.